AN ABELIAN GERBE HAS AN OBJECT OVER ANY NEIGHBOURHOOD ON WHICH THE RELEVANT COHOMOLOGY VANISHES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the local objects, the triviality of the isom torsors, the cohomological obstruction, its vanishing and the correction of the isomorphisms are all written out; the parenthetical that one computation is omitted concerns the verification that the sections form a two-cocycle, a routine cocycle check, and it is preserved verbatim in the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Gerbes and their abelian automorphism sheaves. Stacks, Definition stacks-definition-gerbe and Lemma stacks-lemma-gerbe-abelian-auts. Supplies the local objects of the gerbe and the abelian sheaf of automorphisms.
- Torsors under an abelian sheaf and their triviality. Same chapter, Lemmas lemma-torsors-h1 and lemma-trivial-torsor. Turns the vanishing of the first cohomology on the pieces into isomorphisms between the local objects.
- The Cech-to-cohomology spectral sequence. Same chapter, Lemma lemma-cech-spectral-sequence. Shows that the comparison map from Cech cohomology in degree two to cohomology is injective when the first cohomology of the pieces vanishes.

Source reference. Original lemma 0CJY: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1669–1669. Navigation pointer invoked at proof line 1669; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original definition 06NZ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1694–1694. Navigation pointer invoked at proof line 1694; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03AJ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 291–291. Navigation pointer invoked at proof line 1715; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03AI: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 278–278. Navigation pointer invoked at proof line 1717; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03AZ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1475–1475. Navigation pointer invoked at proof line 1738; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $p : \mathcal{S} \to \mathcal{C}$ be a gerbe over a site whose automorphism sheaves are abelian. Let $\mathcal{G}$ be the sheaf of abelian groups constructed in Stacks, Lemma 0CJY. Let U be an object of $\mathcal{C}$ such that*

1. *there exists a cofinal system of coverings $\{U_i \to U\}$ of U in $\mathcal{C}$ such that $H^1(U_i, \mathcal{G}) = 0$ and $H^1(U_i \times_U U_j, \mathcal{G}) = 0$ for all i, j, and*
2. *$H^2(U, \mathcal{G}) = 0$.*

Then there exists an object of $\mathcal{S}$ lying over U.

Proof. By Stacks, Definition 06NZ there exists a covering $\mathcal{U} = \{U_i \to U\}$ and x_i in $\mathcal{S}$ lying over U_i. Write $U_{ij} = U_i \times_U U_j$. By (1) after refining the covering we may assume that $H^1(U_i, \mathcal{G}) = 0$ and $H^1(U_{ij}, \mathcal{G}) = 0$. Consider the sheaf

$$\mathcal{F}_{ij} = Isom(x_i|_{U_{ij}}, x_j|_{U_{ij}})$$

on $\mathcal{C}/U_{ij}$. Since $\mathcal{G}|_{U_{ij}} = Aut(x_i|_{U_{ij}})$ we see that there is an action

$$\mathcal{G}|_{U_{ij}} \times \mathcal{F}_{ij} \to \mathcal{F}_{ij}$$

by precomposition. It is clear that $\mathcal{F}_{ij}$ is a pseudo $\mathcal{G}|_{U_{ij}}$-torsor and in fact a torsor because any two objects of a gerbe are locally isomorphic. By our choice of the covering and by Lemma 03AJ these torsors are trivial (and hence have global sections by Lemma 03AI). In other words, we can choose isomorphisms

$$\varphi_{ij} : x_i|_{U_{ij}} \longrightarrow x_j|_{U_{ij}}$$

To find an object x over U we are going to massage our choice of these φ_{ij} to get a descent datum (which is necessarily effective as $p : \mathcal{S} \to \mathcal{C}$ is a stack). Namely, the obstruction to being a descent datum is that the cocycle condition may not hold. Namely, set $U_{ijk} = U_i \times_U U_j \times_U U_k$. Then we can consider

$$g_{ijk} = \varphi_{ik}^{-1}|_{U_{ijk}} \circ \varphi_{jk}|_{U_{ijk}} \circ \varphi_{ij}|_{U_{ijk}}$$

which is an automorphism of x_i over U_{ijk}. Thus we may and do consider g_{ijk} as a section of $\mathcal{G}$ over U_{ijk}. A computation (omitted) shows that $(g_{i_0 i_1 i_2})$ is a 2-cocycle in the Čech complex $\check{C}^\bullet(\mathcal{U}, \mathcal{G})$ of $\mathcal{G}$ with respect to the covering $\mathcal{U}$. By the spectral sequence of Lemma 03AZ and since $H^1(U_i, \mathcal{G}) = 0$ for all i we see

that $\check{H}^2(\mathcal{U},\mathcal{G}) \to H^2(U,\mathcal{G})$ is injective. Hence $(g_{i_0i_1i_2})$ is a coboundary by our assumption that $H^2(U,\mathcal{G}) = 0$. Thus we can find sections $g_{ij} \in \mathcal{G}(U_{ij})$ such that $g_{ik}^{-1}|_{U_{ijk}} g_{jk}|_{U_{ijk}} g_{ij}|_{U_{ijk}} = g_{ijk}$ for all i,j,k. After replacing φ_{ij} by $\varphi_{ij} g_{ij}^{-1}$ we see that φ_{ij} gives a descent datum on the objects x_i over U_i and the proof is complete. $\square$